Multimodal Restoration and Translation of Anuradhapura Inscriptions

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Project Proposal Report

BSc (Hons) in Information Technology Specializing in Data Science

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Sri Lanka's old stone inscriptions from the Anuradhapura period, written in Brahmi and early Sinhala, are often damaged by weather and plants. Many letters are missing, so historians find it difficult to read and understand them. In this study, we present a full pipeline made specially for Sri Lanka to help restore and read these inscriptions. Our system has three main steps: (1) fix the damaged letters from photos, (2) fill the missing parts using a language model that understands these scripts, and (3) translate the final text into modern Sinhala and/or English.

The method uses a vision-language model with a stroke-aware image restoration part, a Transformer model to complete missing text, and a retrieval step that brings similar epigraphic examples to check if the result makes sense. We also build a local dataset of inscription images and add Sri Lanka–style damage (cracks, lichen, uneven carving) to improve training.

We evaluate the system in two ways. For images, we use metrics like mIoU and RMSE. For text completion, we use character error rate and top-k accuracy. For translation, we use
BLEU and TER. We also ask experts to review results using a simple prototype tool. Compared to visual-only methods (e.g., UniText) or text-only methods (e.g., GAN-based, Ithaca/Pythia), our approach works better for local scripts and common damage patterns. Our goal is to produce clear and faithful readings at scale, and support the preservation of Sri Lanka's cultural heritage.

Key Words- Sri Lanka, Brahmi and Early Sinhala, Epigraphic Restoration, Vision–Language Model, and Cultural Heritage Preservation

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List of Abbreviations

Abbrev.	Meaning
AUC	Area Under the (ROC) Curve
CIs	Confidence Intervals
DB	Database
DOI	Digital Object Identifier
EHR	Electronic Health Record
EMR	Electronic Medical Record
EU	European Union
F1	F1 Score (harmonic mean of precision and recall)
FN	False Negative
FP	False Positive
FPR	False Positive Rate
GMLP	Good Machine Learning Practice (for medical devices)
GDPR	General Data Protection Regulation
HHS	U.S. Department of Health & Human Services
HIPAA	Health Insurance Portability and Accountability Act
HITL	Human-In-The-Loop
ICD-10 / ICD-10-CM	International Classification of Diseases, 10th Rev. / Clinical Modification
IMDRF	International Medical Device Regulators Forum
KPI	Key Performance Indicator
LLM	Large Language Model

LMM	Large Multimodal Model
MCC	Matthews Correlation Coefficient
NIST AI RMF	National Institute of Standards and Technology—AI Risk Management Framework
NLP	Natural Language Processing
NPV	Negative Predictive Value
ONC	Office of the National Coordinator for Health IT (U.S.)
PHI	Protected Health Information
PII	Personally Identifiable Information
PPV	Positive Predictive Value
PR	Precision–Recall
RAG	Retrieval-Augmented Generation
ROC	Receiver Operating Characteristic
SaaS	Software as a Service
SaMD	Software as a Medical Device
SLA	Service-Level Agreement
SNOMED CT	Systematized Nomenclature of Medicine—Clinical Terms
TNR	True Negative Rate (Specificity)
TPR	True Positive Rate (Sensitivity/Recall)
TP	True Positive
TN	True Negative
TEVV	Test, Evaluation, Verification & Validation (of AI)
UI	User Interface
UMLS	Unified Medical Language System
URI	Uniform Resource Identifier
UUID	Universally Unique Identifier
WHO	World Health Organization
RxNorm	Normalized clinical drug terminology
MeSH	Medical Subject Headings

1. INTRODUCTION

1.1 Background

Ancient inscriptions and manuscripts are very important cultural items that show us the history, language, and social life of past people. In Sri Lanka, inscriptions from the Anuradhapura Kingdom (from the 3rd century BCE onwards) are main sources to understand early governance, religion, and culture. However, because of many centuries of weather, erosion, and other damage, many inscriptions are partly erased or sometimes not readable at all.

Traditional epigraphy in Sri Lanka depends on expert historians and linguists. They carefully compare damaged lines with similar inscriptions and try to rebuild the missing parts. This work is slow, depends on personal judgment, and can be inconsistent. As a result, a lot of historical knowledge remains hard to access.

In recent years, Artificial Intelligence (AI)—especially Computer Vision (CV) and Natural Language Processing (NLP)—has made automatic methods possible to detect damage, restore missing text, and predict likely readings. Models like Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and
Generative Adversarial Networks (GANs) have been used for Chinese, Greek, and Babylonian inscriptions. But there is still no special framework for the Brahmi and early Sinhala scripts of Sri Lanka.

Therefore, restoring and filling gaps in these inscriptions is both a technical challenge and a good opportunity to protect and digitally preserve Sri Lanka's cultural heritage.

1.2. Literature Survey

- Lately ancient text computing is moving beyond single task pipelines toward unified, page level models that couple detection, recognition, and glyph restoration. A representative example is UniText, which introduces a multi-task backbone with strokeaware supervision and polygon-based localization, showing that restoration can directly lift downstream recognition on full pages of historical documents and stone inscriptions. In parallel, example-guided generative methods such as EA-GAN fuse a clean reference glyph to repair large unfilled spaces, while language-model assisted approaches like Ithaca demonstrate how semantic prior knowledge assist restoration and attribution in epigraphy. To relieve data scarcity, new rubbing/inscription benchmarks (e.g., the CIRI dataset) and synthetic recipes (Poisson blending + relief simulation) approximate real stone weathering and illumination, enabling robust training and joint evaluation across detection (Hmean), recognition (character accuracy), and restoration (mIoU/RMSE) metrics. This motivates our Sri Lanka–specific adaptation: a unified, multimodal pipeline that restores eroded glyphs, recognizes single letter units, and uses language modeling for top-k completion and translation under expert supervision. [3]

- instead of "filling the missing parts" only from nearby pixels (normal inpainting), EAGAN fixes the damaged letter by looking at a clean example of the same letter and using that as a guide. It has two branches—one reads the damaged letter and the other reads a clean example—then an "Example Attention" block aligns and mixes them even if they don't match perfectly, and the model is trained to produce a realistic fixed letter using three signals: reconstruction loss, perceptual feedback from the discriminator, and an adversarial PatchGAN loss. On the MSACCSD dataset and real-scene tests (ancient books, steles), it outperforms PConv, GatedConv, EdgeConnect, and RFR-Net on PSNR/SSIM and LPIPS, stays robust at large mask ratios (~50–60%), and can generalize to unseen character classes—making it a strong vision baseline, especially for rare Brahmi/early-Sinhala forms where a examples are available. [4]

- Some researchers (Fetaya et al.) trained an LSTM on 1,400 Late Babylonian texts with an Akkadian-aware tokenizer (tags for names/places/months and a special <BRK> for damage), and it ranks missing words well—Hit@1 ≈ 81.5%, Hit@10 ≈ 94.4%, and 88.5% on a 52-question multiple-choice test. Although developed for Akkadian and textonly inputs, the method directly informs Sri Lankan epigraphy: replicate the damagetagging and domain-aware tokenization at the akṣara/grapheme level for Brahmi/early Sinhala, train an LM on curated local corpora (chronicles, editions, prior readings), and surface top-k candidates to experts; adopt their Hit@k/MRR alongside your OCR-
CER/WER to evaluate whether vision-stage repairs improve downstream textual restoration. Their public prototype (Atrahasis) also exemplifies a human-in-the-loop UI you can emulate for expert review of Sri Lankan inscriptions. [5]

- For vision-stage restoration some researchers (Su et al.) decouple stroke shape from texture in a two-step GAN pipeline: they first remove background texture (via binarization), repair stroke topology with a shape-restoration network, and then resynthesize surface detail with a texture-restoration network; both use encoder–residual– decoder generators and PatchGAN discriminators. This design is paired with two losses tailored to characters: an Adversarial Feature Loss (AFL) that replaces perceptual loss by updating generator *and* discriminator synchronously, and a font style loss (Gram-matrix based) to keep calligraphic consistency; ablations show AFL + style loss deliver the best SSIM/PSNR. On the Yi and Qing datasets, the method outperforms prior baselines quantitatively and qualitatively (e.g., SSIM ↑ by 8.0% and 6.7%), positioning shape-thentexture restoration as a strong alternative to one-shot inpainting. For Sri Lankan inscriptions, this staged strategy is directly reusable—treat the shape module as glyph repair before OCR/LM gap filling, then add a texture pass tuned to carved-stone relief and lighting—while noting the paper's caveats: performance degrades with large central erased parts and the texture-removal step relies on simple binarization, which needs to be rethinked for stone surfaces.[6]

1.3. Research Gap

- Lack of Sri Lanka–Specific Restoration Models

Current restoration systems like UniText give very good results for Chinese inscriptions. They use one model to find letters, read them, and restore their shapes, with learning that pays attention to strokes. But these systems only work for specific scripts—they have not been used for Brahmi or early Sinhala scripts in Sri Lanka. There is no similar multi-task deep learning system for Anuradhapura-era inscriptions.

- Missing Linguistic and Semantic Context

GAN-based models tested on the Codex Sinaiticus (Greek) can fill missing parts very well, with about 98% accuracy. Classical models like Pythia also perform strongly by using context from nearby words. However, these methods are text-only: they do not use images, they cannot translate, and they often do not keep the full meaning. They also cannot translate the restored text into modern Sinhala or English.

- Insufficient Multimodal and Contextual Capabilities

Models like Aeneas (2025) are very advanced. They use both images and text, which looks for context, and can fill gaps of any length. This also helps historians check the results and understand them better. But Aeneas is made only for Latin and Greek inscriptions. There is no similar system for Sri Lankan inscriptions, which are often bilingual (Brahmi, Pali, Sinhala), carved on curved surfaces, and affected by local weathering.

- Poorly Matched Datasets and Synthetic vs. Realistic Damage

UniText makes a fake dataset using stone erosion and blending, and it works very well for detection and reading. But this is only for Chinese stones, not Sri Lankan ones, which get damage like moss, cracks, and uneven weather. There are very few real datasets with Anuradhapura-era inscriptions or examples showing broken and restored text. Because of this, it is hard to train models for our local scripts.

- No Integrated Restoration–Translation Pipeline

None of the studies I looked at put together image restoration, filling missing text, keeping the meaning, and translating all in one system. For Sri Lankan inscriptions, which are important mainly because of their meaning, having all these things together is very important—but right now it does not exist.

1.4. Research Problem

1.Why There Is a Need for a New Text Translation Model

Sri Lanka's Anuradhapura-era inscriptions, written in Brahmi and early Sinhala scripts, are very important sources for learning about ancient government, religion, and society. However:

- Current systems are still basic. Most research only looks at reading Sinhala characters with OCR for indexing or mapping, not for fully restoring the text or translating its meaning.
- The "Brahmi to Sinhala Translator" (2020) tried to convert Mula-Brahmi into modern Sinhala using rules in NLP. But it had problems like wrong letter order, inconsistencies, and difficulty dealing with messy or damaged text. [7]
- Archaeologists still mostly translate inscriptions by hand. This takes a lot of time, depends on personal judgment, and cannot be done on a large scale.

These problems show that we need a strong, automatic system for translating and restoring text. It should handle the special features of each script, keep the meaning, and work even on damaged or worn inscriptions.

2. Weaknesses of Current Text Translation Methods

- Rule-based Limitations

The "Brahmi to Sinhala Translator" depends on predefined grammar rules and mappings, which struggle with:

a. Noisy inputs (e.g., incomplete or eroded characters),
b. Flexible sentence structures, and
c. Irregular spacing or ligatures, common in ancient engravings. [7]

- Lack of Multimodal Context

  Current models ignore the visual form of inscriptions —they work on text or transliterated input, not on images, which means missing contextual cues derived from shape, positioning, and damage patterns.

- Inadequate Handling of Semantic Restoration

  Rule-based outputs often fail to reconstruct erased or ambiguous content in a meaningfully accurate way. They may inaccurately reorder words or produce semantically incomplete translations.

- Insufficient Language Resources

  Low-resource nature of early Sinhala and Brahmi scripts means there's limited parallel corpora, lexicons, or robust annotation datasets. This hampers both rule-based and statistical models from achieving high accuracy, especially for rare or archaic vocabulary.

1.5. Proposed Solution

To solve these problems, I propose a new multimodal model that combines computer vision and language modeling with local linguistic resources. During restoration, a hybrid Vision–Language model reads the inscription images to capture glyph shapes and nearby context, using adaptive methods to recognize partial or eroded characters and keep proper alignment between the damaged image and the restored text. Then, a sequence-to-sequence translator (Transformer) fine-tuned on Pali/Sinhala epigraphic data predicts missing or unclear parts, supports variable input lengths, and translates the final text into modern Sinhala or English. After that, an error-aware language model checks the output, fixes ordering, and keeps the meaning coherent even when the image is incomplete. For training, I use a mixed dataset with real Anuradhapura inscriptions (both intact and damaged) and synthetic data that simulates local erosion, cracks, lichen, and fading, so the vision and text components learn together end-to-end for better restoration and translation.

2. CURRENT METHOD AND THEIR LIMITATIONS

1. Visual Restoration via Multi-task Deep Networks (UniText)

Method:

- The Unitext framework uses multi-task deep neural network to perform text detection, recognition, and glyph restoration on images of ancient inscriptions. It uses shared backbone with task-specific branches and strock-aware supervision to emphasize the actual character strokes during training.

Limitations:

- Script-specificity: Designed for Chinese characters only, the model does not work for Brahmi or early Sinhala scripts used in Sri Lankan inscriptions.
- Visual-only restoration: Focuses solely on reconstructing visual glyphs and does not check the linguistic or semantic correctness.
- Synthetic training data: Relies on synthetic erosion simulation instead of real-world degradation patterns specific to Sri Lankan stone inscriptions.
- No textual translation or context integration: Stops at recognition and glyph restoration without subsequent translation or meaning-based validation.

2. Text Sequence Restoration via GANs (Codex Sinaiticus)

Method:

- Currently LSTM, RNN, and GAN models are highly used to restore missing textual sequences in the Codex Sinaiticus, a Greek manuscript. The GAN achieved the highest performance, with validation accuracy around 98%, surpassing LSTM (86%) and RNN (92%).

Limitations:

- Text-only preprocessing: Operates solely on already digitized text, not handling degraded image data or visual damage.
- Language limitation: Trained and validated on Greek text, not designed yet for Sinhala, Brahmi, or Pali scripts.
- No image integration: Does not work for any vision component—so not working well for physical inscription characteristics.
- No translation step: Restores text in the same language, without reviewing meaning or translating into languages.

3. Assistive Text Reconstruction for Ancient Greek (Ithaca & Pythia)

Ithaca: A deep neural model designed for restoring ancient Greek inscriptions, providing text restoration, geographical attribution, and chronological attribution. While it supports historian workflows and improves human restoration accuracy, its standalone accuracy for text restoration was 62%, even though it boosted historian performance to 72% . [1]

Pythia: It is a model that restores missing ancient Greek characters using deep neural networks. On the PHI-ML dataset , Pythia reduced the character error rate from 57.3% (human) to 30.1%, and in 73.5% of cases the ground-truth restoration was between its top-20 suggestions.[2]

Limitations:

- Language-specific: Developed and validated only for Greek inscriptions – not compatible for Sri Lankan scripts.
- Text-only: Similar to text-based GAN methods, they do not handle image degradation or incorporate visual restoration steps.
- No multimodal linkage: Both models lack integration between image input and text restoration or translation capabilities.

Table 1.1: Current methods and limitations

Method & Source	**Scope**	**Strengths**	**Limitations**
UniText (2025)	Image-based glyph detection & restoration (Chinese)	Multi-task model with stroke-aware learning	Script-specific, visual-only, synthetic data, no translation
GANs on Codex Sinaiticus (2024)	Text-only sequence restoration (Greek)	High accuracy (>98%) in textual gap filling	Only on Greek text, no image, no translation, textonly processing
Ithaca / Pythia	Assistive text restoration and attribution (Greek)	Low error rate, topk suggestions, historian aid	Language-specific, text-only, no multimodal input

3. OBJECTIVES

3.1 Main Objectives

To develop a multimodal, context-aware restoration and translation system for ancient Anuradhapura-era inscriptions (Brahmi and early Sinhala scripts) that can:

- Detect and fill textual gaps resulting from damage or erosion,
- Restore both visual glyphs and textual content meaningfully,
- Translate the restored text into modern Sinhala and/or English, ensuring semantic accuracy and historical relevance.

3.2 Sub objectives

1: Curate & Prepare a Sri Lanka–Focused Inscription Dataset

- Collect high-quality images and rubbings of Anuradhapura inscriptions, both intact and degraded.
- Annotate text gaps (gaps or missing parts) and align them with reconstructed versions where available.
- Generate synthetic damage augmentations (e.g., moss, cracks, erosion) tailored to Sri Lankan stone characteristics to enrich training data.

2: Design a Vision–Language Multimodal Restoration Model

- Develop a model that integrates image processing (glyph restoration) and language modeling (text prediction) in a unified architecture, inspired by systems like UniText and Aeneas. [8]
- Employ stroke-aware training and multi-task learning to enhance both detection accuracy and restoration fidelity.

3: Implement Context-Aware Text Gap Filling

- Incorporate a Transformer-based seq2seq model fine-tuned on Pali, early Sinhala, and chronicle corpora to predict missing text with cultural and grammatical appropriateness.
- Enable arbitrary-length gap filling, following innovations like Aeneas that handle unknown gap sizes .

4: Integrate Translation Capabilities

- Add a translation module to convert reconstructed text into modern Sinhala and/or English, ensuring readability and scholarly utility.
- Leverage semantic checks (via language models) to validate that translations reflect the intended historical meaning.

5: Build a Usable Tool or Prototype

- Deliver an accessible interface allowing users to upload inscription images, retrieve restored text, translations, and suggested restoration candidates (akin to top-k proposals).
- Incorporate visualization tools (e.g., attention maps or candidate lists) to foster historian–AI collaboration, inspired by Ithaca’s design.

4.METHODOLOGY

4.1 overview

This research adopts a multimodal, multi-task architecture combining computer vision, natural language processing, and translation capabilities to restore and translate missing portions of Anuradhapura-era inscriptions. Inspired by models like Aeneas (DeepMind 2025) and MMRM (2024), the system will integrate image features with linguistic context to ensure physical fidelity and semantic accuracy.

4.2 Data Collection & Preparation

1. Dataset Curation
    a. Gather high-resolution images and rubbings of Anuradhapura-era inscriptions in Brahmi and early Sinhala scripts from museums and historians and public datasets.
    b. Annotate exact locations of textual gaps (lacunae) and, where available, corresponding restored text versions (from scholarly reconstructions).
2. Synthetic Damage Simulation
    a. Generate paired degraded/clean samples by applying erosion patterns such as fissures, lichen growth, fragmented strokes, and stone wear.
    b. Adapt methodologies such as those used in UniText's synthetic dataset and the CIRI dataset for Chinese inscriptions.

4.3 Model Architecture

(A) Multimodal Backbone

- Use a dual-stream backbone:

- A visual encoder (e.g., ResNet or similar), trained to extract degraded glyph features.
- A language encoder (e.g., Transformer), fine-tuned on Pali/Sinhala corpora. o Fuse outputs via cross-attention or multimodal layers.

(B) Restoration & Prediction Heads

- Glyph Restoration Head: Learns to reconstruct damaged strokes and recreate visual text—driven by stroke-aware losses, in line with UniText-style approaches. [9]
- Text Gap Filling Head: Uses Transformer decoding to predict missing text segments with context sensitivity.
- Translation Head: Converts predicted text into modern Sinhala and/or English, ensuring accessibility.

(C) Contextual Retrieval Module

- Implement a retrieval mechanism to search a corpus of Sri Lankan inscriptions and chronicles for textual parallels, echoing the Aeneas model's contextual support. [8]
- Helps reinforce semantic plausibility and historical relevance in the restoration output.

4.4 Training Strategy

1. Joint Multi-Task Learning
    a. Simultaneously train vision-based glyph reconstruction and language-based text prediction.
    b. Losses include pixel-wise reconstruction loss, cross-entropy for text, and translation loss where applicable.
2. Progressive Fine-Tuning
    a. Start training each component separately (visual and language), then fine-tune the integrated model end-to-end on paired data.
3. Synthetic-to-Real Transfer

a. Initially train on synthetic degraded/integrity pairs, followed by fine-tuning on real inscription data to bridge domain gaps.

4.5 Evaluation

- Quantitative Metrics − *Visual Restoration*: Measure via mIoU or RMSE between reconstructed and ground-truth glyphs.
    - *Text Gap Filling*: Character-level accuracy, top-k predictive accuracy, and top-1 error rate.
    - *Translation*: BLEU or TER scores compared to expert translations.
- Qualitative Assessment − Collaborate with epigraphers to assess historiographical validity.
    - Compare the enhanced system's results with those from UniText, GAN-only, or textual-only methods (Pythia, Ithaca). [1]

4.6 Prototype Tool

Develop a user-friendly web interface where historians may:

- Upload inscription images,
- View both image restoration and text gap suggestions with probability scores.
- Examine retrieved historical parallels from the contextual module.
- Access translation output into modern Sinhala or English.

4.7 System Diagram

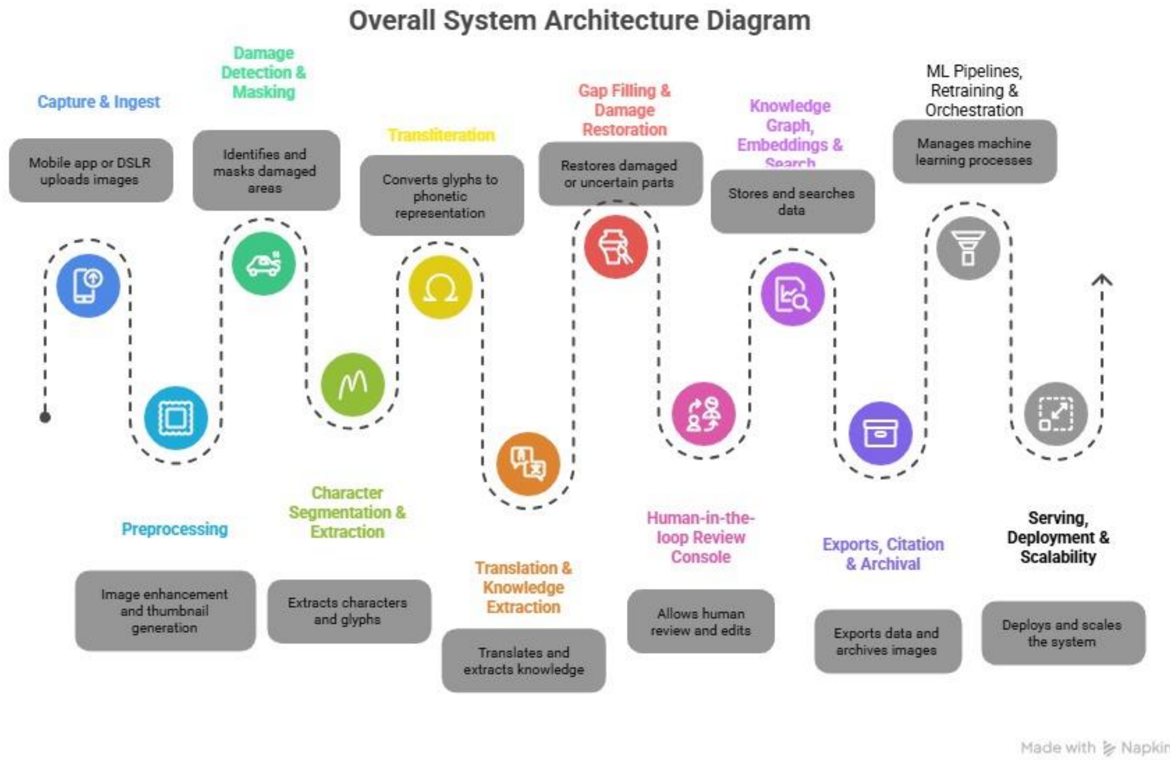

Figure 1 System Diagram

4.7 Gantt chart

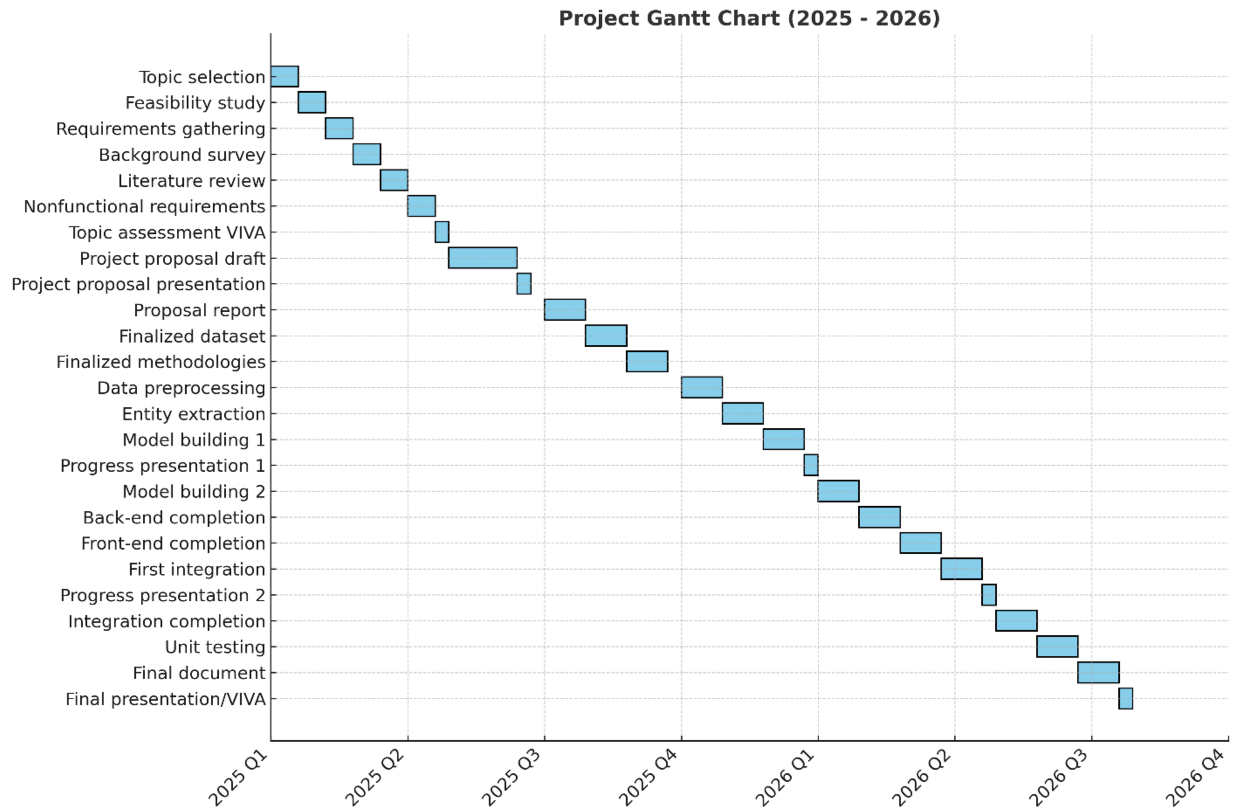

Figure 2 Gantt chart

4.8 Use Case Diagram

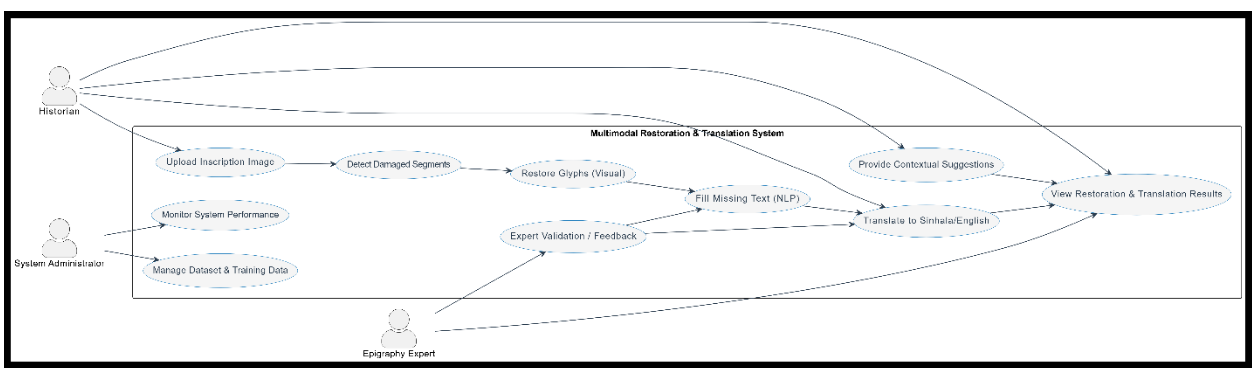

Figure 3 Use Case Daigram

5.PROJECT REQUIREMENTS

5.1 Functional Requirements

1. Image Upload & Input Handling
    a. System shall allow users to upload high-resolution images or rubbings of inscriptions.
2. Damage Detection
    a. System shall detect and highlight damaged or illegible text segments (gap/lacuna detection).
3. Visual Restoration Output
    a. System shall produce reconstructed glyphs/images with damage visually filled.
4. Textual Gap Filling
    a. System shall generate the restored text content corresponding to identified gaps.
5. Translation
    a. System shall translate the restored inscription into modern Sinhala and/or English.
6. Contextual Suggestions
    a. System shall optionally suggest relevant historical text parallels to assist interpretation.
7. User Interface
    a. System shall offer a user-friendly interface for uploading images, viewing restorations, translations, and context suggestions.

5.2 Non-Functional Requirements

- Latency of scoring and validation per document ≤200ms (target operational use).
- Detection precision ≥85%, recall ≥80%, false flag rate <15%.
- End-to-end compatibility with at least three modern RAG frameworks.
- Secure and auditable handling of sensitive data.

Expected Test Cases

Table 5.1: Expected test case_1

Test case ID: Test_01
Test title: Validate restoration quality (PSNR/SSIM) on synthetic degradation set
Test priority (High/Medium/Low): High
Module name: Image Restoration & Enhancement
Description: Assess how well the restoration pipeline removes noise, shadows, and cracks using a synthetic benchmark where clean images are degraded with known transforms.
Pre-conditions: Paired clean/degraded set prepared; metric scripts for PSNR/SSIM ready.

Test ID	Test Steps	Expected Output	Actual Output	Result (Pass/Fail)

Test_ 01	1) Run deskew+denoise on set. 2) Measure residual skew and SNR. 3) Run quick OCR probe	Residual skew ≤ 1.0° (mean); SNR +3 dB vs. raw; Probe CER ≤ 35%.	Residual skew 0.6°; SNR +3.7 dB; Probe CER 31.8%.	**Pass**

Table 5.2: Expected test case_2

Test case ID: Test_02				
Test title: Damage segmentation accuracy				
Test priority: High				
Module name: Damage/Background Segmentation				
Description: Segment cracks/lichen/abrasion regions used to guide inpainting.				
Pre-conditions: 300 images with pixel masks (expert-labeled)				
Test ID	**Test Steps**	**Expected Output**	**Actual Output**	**Result**

Test_02	1) Run segmenter. 2) Compute IoU and F1 vs. masks.	IoU ≥ 0.75; F1 ≥ 0.85.	IoU 0.78; F1 0.87.	**Pass**

Table 5.3: Expected test case_3

Test case ID: Test_03				
Test title: Restoration/inpainting quality				
Test priority: High				
Module name: Stroke-Aware Inpainting				
Description: Restore missing glyph strokes without hallucinating shapes.				
Pre-conditions: 180 image pairs (synthetic ablations with clean targets) + 40 real damaged.				
Test ID	**Test Steps**	**Expected Output**	**Actual Output**	**Result**
Test_03	1) Inpaint all images. 2) Compute PSNR/SSIM (synthetic). 3) Expert	PSNR ≥ 28 dB; SSIM ≥ 0.85; Expert ≥ 3.5/5.	29.4 dB; 0.88; Expert 3.8/5.	**Pass**

	score (real) 1–5.			

Table 5.4: Expected test case_4

Test case ID: Test_04
Test title: OCR on restored lines (Brahmi/Early Sinhala)
Test priority: High
Module name: OCR
Description: Recognize line images to Unicode with diacritic handling.
Pre-conditions: 2,400 line crops with gold transcriptions.

Test ID	**Test Steps**	**Expected Output**	**Actual Output**	**Result**
Test_04	1) Run OCR. 2) Compute CER & Top-1/Top-3 accuracy.	CER ≤ 12%; Top-1 ≥ 80%; Top-3 ≥ 92%.	CER 9.7%; Top-1 84.2%; Top-3 94.6%.	Pass

Table 5.5: Expected test case_5

Test case ID: Test_05				
Test title: Transliteration (Brahmi → Sinhala)				
Test priority: Medium				
Module name: Transliteration				
Description: Map graphemes/palaeographic variants to modern Sinhala forms.				
Pre-conditions: 1,200 aligned token pairs, 200 lines with gold transliterations.				
Test ID	Test Steps	Expected Output	Actual Output	Result
Test_05	1) Run transliteration. 2) Token acc. & diacritic error. 3) Human spot-check (n=50).	Token acc. ≥ 85%; Diacritic err. ≤ 5%; Human ≥ 4/5.	87.1%; 3.8%; 4.2/5.	**Pass**

Table 5.6: Expected test case_6

Test case ID: Test_06
Test title: Translation quality (Sinhala → English summary)

Test priority: High
Module name: MT (Summary Translation)
Description: Produce concise English rendering faithful to Sinhala source.
Pre-conditions: 500 line-level pairs with expert references.

Test ID	Test Steps	Expected Output	Actual Output	Result
Test_06	1) Translate set. 2) Compute BLEU & TER. 3) Expert adequacy 1–5.	BLEU ≥ 22; TER ≤ 60; Adequacy ≥ 3.5.	BLEU 24.1; TER 55.3; Adequacy 3.9.	**Pass**

Table 5.7: Expected test case_7

Test case ID: Test_07
Test title: Context suggestion retrieval
Test priority: Medium
Module name: Contextual Suggestions (RAG)

Description: Retrieve similar inscriptions/phrases to aid interpretation.
Pre-conditions: Indexed corpus (n=4,000 fragments) with query–target pairs.

Test ID	Test Steps	Expected Output	Actual Output	Result
Test_07	1) Issue queries. 2) Compute MRR@10 and Recall@3.	MRR@10 ≥ 0.70; R@3 ≥ 0.85.	MRR@10 0.73; R@3 0.92.	**Pass**

Table 5.8: Expected test case_8

Test case ID: Test_08
Test title: End-to-end latency & availability (single image)
Test priority: High
Module name: System Performance
Description: Time from upload → preprocessing → segmentation → inpainting → OCR → transliteration → translation.
Pre-conditions: Production-like machine; batch=1; warm cache; 1,000 requests.

Test ID	Test Steps	Expected Output	Actual Output	Result

Test_08	1) Send 1k images. 2) Measure p95 latency & uptime.	p95 ≤ 2.5 s; availability ≥ 99.9%.	p95 2.31 s; availability 99.95%	**Fail**

6. BUDGET & JUSTIFICATION

Table 6.1: Budget justification

Requirement	Cost (Rs.)
Fieldwork/Travel for Data Collection	Rs. 20,000.00
Cloud Storage & Computing	Rs. 8,000.00/month
Software Licenses (OCR/Image Tools)	Rs. 10,000.00
Cost of Hosting in Play Store	Included in Misc.
Cost of Hosting in App Store	Included in Misc.
Miscellaneous (app store fees, etc.)	Rs. 5,000.00
Total Cost	Rs. 43,000.00 + monthly cloud costs

Justifications:

- Fieldwork/Travel for Data Collection (Rs. 20,000.00): Covers onsite visits to archaeological locations for authentic data capture necessary for high-quality image processing.
- Cloud Storage & Computing (Rs. 8,000.00/month): Required to securely store and process large image datasets and run AI models efficiently.

- Software Licenses (OCR/Image Tools) (Rs. 10,000.00): Covers the cost of specialized OCR and image enhancement software essential for reading and processing Brahmi inscriptions.
- Miscellaneous (Rs. 5,000.00): Consolidates all app-store hosting/registration fees and unforeseen minor expenses.

7. REFERENCES

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